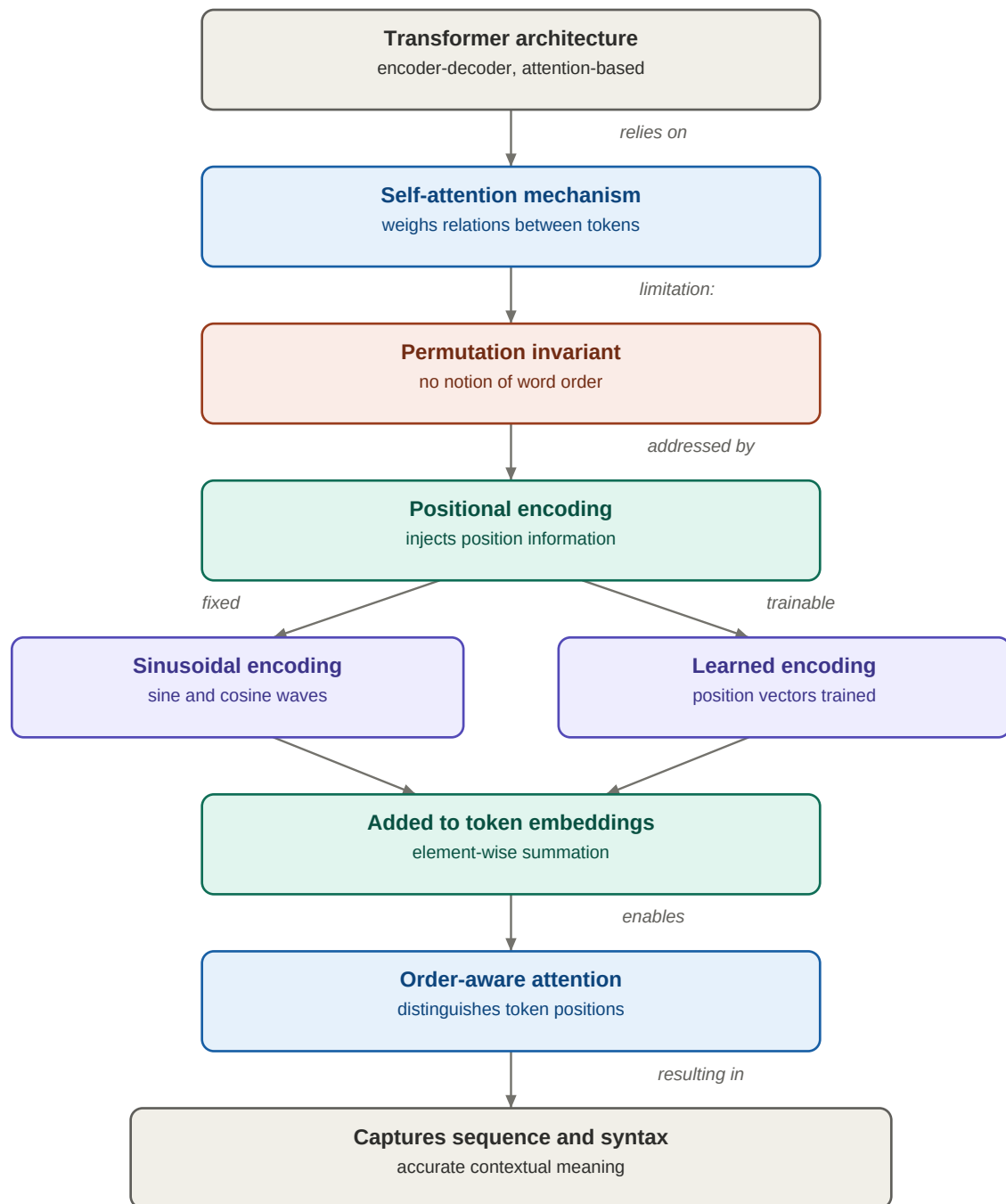


Concept map answers

Role of Positional Encoding in Transformer Architecture



Comparison and justification

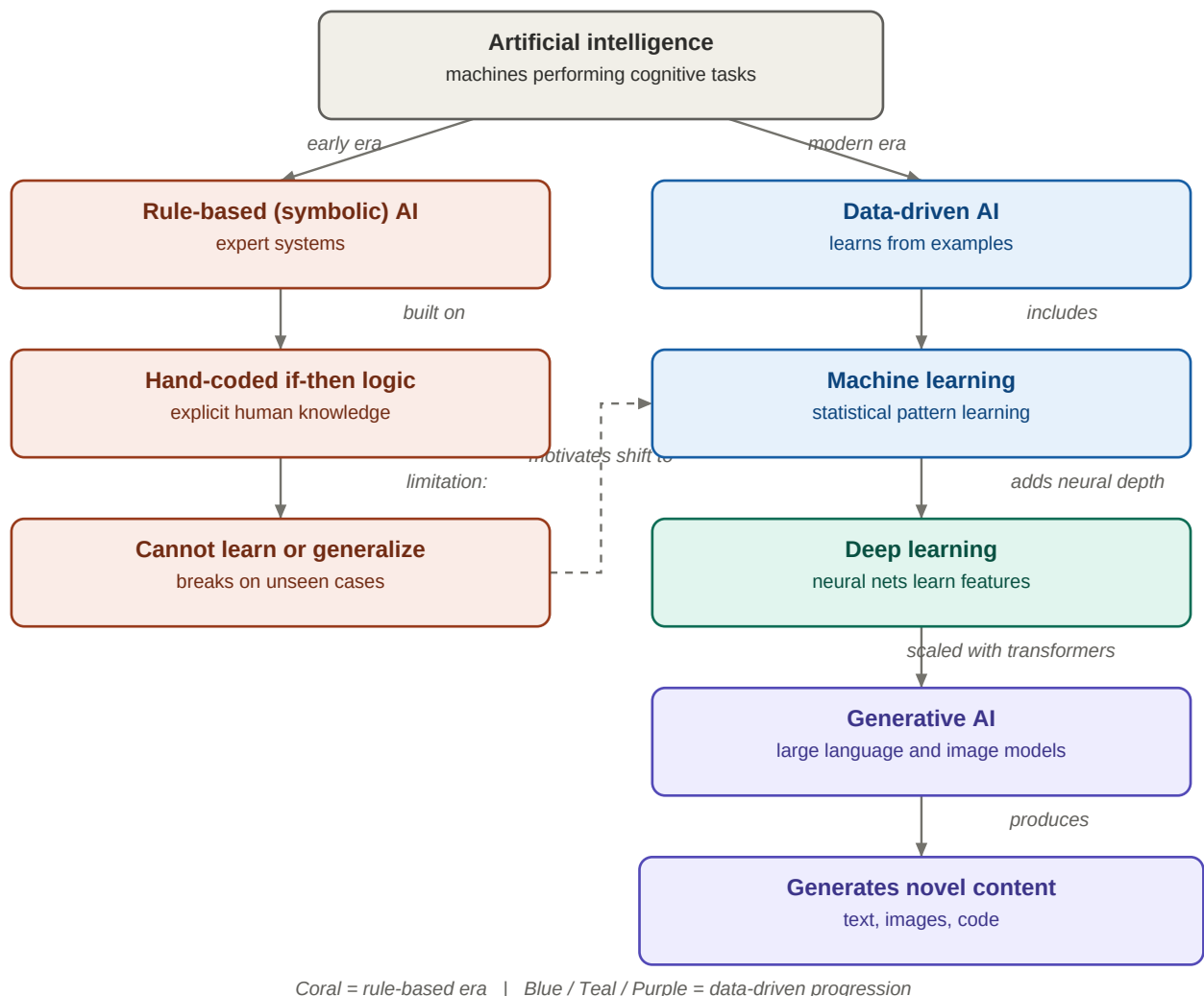
Self-attention computes weighted relationships between all tokens simultaneously, which is what makes it fast — but that same parallelism is why it has no built-in sense of order, unlike a recurrent network that processes tokens one after another. Positional encoding fixes this without giving up parallelism: it adds a position vector to each token's embedding before attention runs.

Sinusoidal vs. learned encoding: sinusoidal encoding uses fixed sine and cosine functions of varying frequency, so it needs no extra training and can generalize to sequence lengths longer than any seen

during training. Learned encoding instead trains a position vector per slot, which can fit the training data more precisely but cannot represent positions beyond the maximum length it was trained on. The trade-off is generalization versus flexibility.

Justification of linkage: both methods feed into the same step — summation with token embeddings — because position information must travel through every downstream attention layer alongside meaning; if it were applied only once and then discarded, deeper layers would lose track of order.

Evolution of Artificial Intelligence — Rule-Based Systems to Generative AI



Comparison and justification

Rule-based AI vs. machine learning: rule-based systems encode human expertise directly as logic (for example, medical diagnosis expert systems of the 1970s–80s), so they are transparent and easy to audit but brittle — any case outside the programmed rules fails outright. Machine learning instead infers patterns statistically from labeled data, trading transparency for adaptability; it handles unseen cases probabilistically instead of failing on them. This is why the dashed cross-link in the map connects the rule-based branch's core limitation directly to the start of the data-driven branch — the shift was a direct response to that failure mode, not an incidental one.

Machine learning vs. deep learning: classical machine learning still needs a human to hand-design the input features (for example, choosing which pixel statistics matter for image recognition), whereas deep learning's layered neural networks learn the features themselves directly from raw data, removing that manual bottleneck.

Deep learning vs. generative AI: earlier deep learning was largely discriminative — trained to classify or predict a label for a given input. Generative AI, enabled by the transformer architecture and massive training scale, is trained instead to model the full distribution of the data, so it can produce new, original outputs rather than only judge existing ones.